

4. Hypothesis tests

What students need to learn:

Population, census and sample. Sampling unit, sampling frame.

Students will be expected to know the advantages and disadvantages associated with a census and a sample survey.

Concepts of a statistic and its sampling distribution.

Concept and interpretation of a hypothesis test. Null and alternative hypotheses.

Use of hypothesis tests for refinement of mathematical models.

Critical region.

Use of a statistic as a test statistic.

One-tailed and two-tailed tests.

Hypothesis tests for the parameter p of a Binomial distribution and for the mean of a Poisson distribution.

Students are expected to know how to use tables to carry out these tests. Questions may also be set not involving tabular values or requiring the use of normal approximations.
